

BSE4203: Software Engineering Standards and Ethics

Submission Date: Friday, 1st May, 2026

Presentation Dates: Monday 4th-Tuesday 5th May, 2026

Capstone Title: "Ethics and Standards Compliance in Final Year Software Engineering Projects"

Overview:

This capstone project integrates software engineering ethics and standards into your existing final year projects. You will work in your final year project groups and systematically apply ethical considerations and software standards to your projects over 10 weeks, with clear weekly deliverables. In each week, your group will analyze, implement, and document how your software aligns with professional ethics, legal compliance, and internationally recognized software standards.

Capstone Rules (applies every week)

Group size: 4 students

Case Study: Your Final Year Software Engineering Project (FYP)

Tooling: ClickUp collaboration space per group

Weekly work

- Every student in the group must have **assigned tasks** in ClickUp (with due dates, assignees, and status).
- Every week must end with a small **Weekly Report** and an **Evidence Pack**.
- The project management Tasks must be assigned to a different individual every week.
- Each deliverable must include a link to the ClickUp space and screenshots/export evidence.

ClickUp setup requirement (part of Week 1)

- Each group must create ClickUp Space named: BSE4203-GroupX-FYP-<ProjectShortName>
- Invite the lecturer: kougaus@gmail.com with admin/member access as you prefer.
- Use a consistent structure (suggested below).

Recommended ClickUp Structure (This must be standard for all groups)

Space: BSE4203-GroupX-FYP-ProjectShortName

Folders (10): Week23-28 ... Week10

Each folder contains:

- **List A: Tasks (Individual)** (every task assigned to a member)
- **List B: Deliverables** (upload/link final PDF/word document and evidence)
- **List C: Weekly Report** (template entry)

Mandatory ClickUp fields

- Assignee, Due date, Priority, Status, Description, Attachments, Checklist
- Add tags: Ethics, Standards, ISO12207, ISO15288, IP, Cyberlaw, Quality, Security, SPI

10-Week Capstone Plan (Deliverable-driven)

Week 1 (23rd -27th February, 2026)— Capstone Kickoff and Project Context Baseline

Indicative content covered: Definition of SE standard; Professional codes intro; obligations overview

Recommended Group Reading

- Reynolds (Ethics in IT): intro chapters on ethics/professional responsibility
- Quinn: Intro and professional ethics
- Skim: ISO/IEC/IEEE 24765 definition terms (standards, process, lifecycle)

Individual tasks (each member)

1. Member 1: Write project overview (problem, users, scope, stakeholders)
2. Member 2: Create system context diagram + stakeholder map
3. Member 3: Create risk brainstorm list (harm/security/privacy/IP)
4. Member 4: Setup ClickUp space structure and invite lecturer

Deliverables (end of Week 1)

1. **ClickUp Evidence Pack** (screenshots: space, members, tasks assigned, lecturer invited)
2. **Project Context & Stakeholder Baseline (2–3 pages)** including:
 - project description, goals, major features and usecases
 - stakeholder map (who is affected and how)
 - initial list of ethical and standards concerns (at least 10 items)
3. **Weekly Report #1:** Includes each member's tasks and outcomes plus next week (Week 2) plan

Week 2 (2nd -6th March, 2026) — Professional Codes of Ethics + “Who are we obligated to?”

Indicative content covered: Professional codes; obligations; where obligations come from
Recommended Reading

- Reynolds/Quinn sections on professional codes
- ACM/IEEE Software Engineering Code of Ethics (principles and examples)

Individual tasks

1. Member 1: Map **ACM/IEEE principles** to the Final Year Project context (minimum 6 mappings)
2. Member 2: Identify key obligation conflicts (e.g., client vs public vs employer)
3. Member 3: Draft “Ethics Charter” for the team (commitments and do/don’t)
4. Member 4: Create ClickUp checklist template for weekly ethics compliance

Deliverable (end of Week 2)

1. **Ethics Charter (1–2 pages)** referencing ACM/IEEE principles
2. **Obligations Map (a table):** stakeholders × obligations × risks
3. **Weekly Report #2 + ClickUp evidence** (task completion)

Week 3 (9th -13th March, 2026) — Harm in Software Engineering + Ethical Analysis Skills

Indicative content covered: Harm; ethical implications that incur harm; constructive habits of mind

Recommended Reading

- Reynolds/Quinn chapters on harm, risk, responsibility
- Case readings: online harms, misuse, unintended consequences

Individual tasks

1. Member 1: Identify 5 harm scenarios (user harm, org harm, societal harm)
2. Member 2: Do severity/likelihood scoring + propose mitigation for each harm
3. Member 3 and Member 4 : Write “Ethical Decision Log” template (Issue → Options → Decision → Justification → Evidence)

Deliverable (end of Week 3)

1. **Harm & Mitigation Register** (minimum 5 harms; scored + mitigations)
2. **Ethical Decision Log v1** with at least **2 filled entries** from your project
3. **Weekly Report #3**

Week 4 (16th -20th March, 2026) — Doing Good: Ethical Design Choices + Controversies (Privacy/Harassment)

Indicative content covered: good life; acting ethically; privacy; online harassment controversy
Recommended Reading

- Quinn/Reynolds privacy chapters
- One privacy/online harassment case study relevant to your domain (edtech, health, business, finance, etc.)

Individual tasks

- Member 1: Draft privacy-by-design goals for your Final Year Project (minimization, consent, access)
- Member 2: Identify harassment/misuse possibilities (even if not social media platform) and safeguards
- Member 3: Create user-facing policy drafts (privacy notice + acceptable use summary)
- Member 4: Define monitoring/complaint handling workflow (report abuse/data issue)

Deliverable (end of Week 4)

1. **Ethical Feature Review Report (2–4 pages):**
 - privacy-by-design decisions
 - misuse/harassment risk and controls
2. **A short Draft Privacy Notice (short) and Acceptable Use Summary**
3. **Weekly Report #4**

Week 5 (23rd -27th March, 2026) — Intellectual Property (IP) + Open-Source Compliance

Indicative content covered: Intellectual property
Reading

- Reynolds/Quinn IP chapters
- Basic overview of OSS licensing (MIT/Apache/GPL concepts)

Individual tasks

1. Member 1: List all third-party libraries/components used (SBOM-style list)
2. Member 2: Identify licenses and obligations (attribution, distribution, copyleft risks)
3. Member 3: Draft IP ownership plan (team, supervisor, university, client—assumptions noted)
4. Member 4: Draft a short “No-plagiarism and code provenance” procedure

Deliverable (end of Week 5)

1. **IP & OSS Compliance Report (2–3 pages)** including:
 - component inventory + license obligations
 - compliance actions (notices, attributions, replacements if needed)

2. Code Provenance Checklist
3. Weekly Report #5

Week 6 (30th March-3rd April, 2026) — — Cyber Law and Security (Legal + Technical Controls)

Indicative content covered: Cyber law and security
Reading

- Reynolds/Quinn cyberlaw/security sections
- A short national/sector legal note relevant to your domain (e.g., data protection principles) (*You can cite generally without deep legal detail*)

Individual tasks

1. Member 1: Create a threat model that includes (top threats + assets + attackers)
2. Member 2: Draft security requirements (auth, access control, logging)
3. Member 3: Draft incident response mini-plan (detect → respond → recover → report)
4. Member 4: Create security test checklist (Static Application Security Testing (SAST)/Dynamic Application Security Testing (DAST)/basic tests)

Deliverable (end of Week 6)

1. **Security & Cyberlaw Compliance Pack**
 - threat model summary
 - security requirements
 - incident response mini-plan
2. **Weekly Report #6**

Week 7 (6th -10th April, 2026) — Evolution of Standards + ISO/IEC 12207 (2008/2017) Applied

Indicative content covered: evolution of standards; ISO/IEC 12207
Reading

- Schneider & Berenbach (2013) (focus on why standards matter; requirements standardization)
- ISO/IEC 12207 overview (2008/2017): lifecycle processes concept (You can use the lecture summaries to guide you)

Individual tasks

1. Member 1: Map your Final Year Project activities to **12207 process groups** (high-level)
2. Member 2: Identify missing process areas (e.g., configuration management, QA)
3. Member 3 and Member 4: Propose a lightweight process improvement plan for your team

Deliverable (end of Week 7)

1. **ISO/IEC 12207 Mapping Table** (process → what we do → evidence)
2. **Process Gap Analysis + Improvement Actions**
3. **Weekly Report #7**

Week 8 (13th -17th April, 2026) — ISO/IEC 15288 (Systems Perspective) + Obligations & Stakeholders

Indicative content covered: ISO/IEC 15288; obligations revisited

Reading

- ISO/IEC 15288 overview (system life cycle; stakeholders; requirements; verification/validation)

Individual tasks

- Member 1: Write system boundaries + external interfaces
- Member 2: Derive stakeholder requirements (non-functional + operational)
- Member 3: Define verification vs validation plan for any 2 key requirements
- Member 4: Update the obligations map with systems-level impacts (safety, misuse)

Deliverable (end of Week 8)

1. **15288 Systems View Pack**
 - system boundary + interfaces
 - stakeholder requirements summary
 - V&V plan excerpt
2. **Weekly Report #8**

Week 9 (20th April-24th April, 2026) — Other Standards: ISO/IEC/IEEE 24765 + ISO 25000 (Quality Model)

Indicative content covered: 24765 terminology; ISO 25000 quality

Reading

- ISO/IEC/IEEE 24765 glossary concepts (terms used in your report)
- ISO 25000 quality characteristics (functional suitability, reliability, usability, security, etc.)

Individual tasks

1. Member 1: Select 5 quality attributes relevant to your Final Year Project and define metrics
2. Member 2: Create a mini quality evaluation plan (how to measure each)
3. Member 3: Create a traceability matrix excerpt (requirements → tests)

4. Member 4: Compile glossary of 10 key standards terms used consistently

Deliverable (end of Week 9)

1. **ISO 25000 Quality Plan (2–3 pages):** attributes + metrics + evaluation approach
2. **Requirements Traceability Matrix (RTM) excerpt** (at least 8 requirements mapped to tests)
3. **Glossary (24765 terms used)**
4. **Weekly Report #9**

Week 10 (27th April -1st May, 2026) — Relationships Between Standards + Final Compliance & Ethics Portfolio

Indicative content covered: relationships among standards; acting ethically; course integration

Individual tasks

1. Member 1: Build “standards relationship map” (12207 ↔ 15288 ↔ 25000 ↔ ethics/IP/security)
2. Member 2: Compile final compliance evidence pack (screenshots, logs, artifacts)
3. Member 3: Finalize ethics decision log (minimum 5 entries across project timeline)
4. Member 4: Prepare final presentation + submission packaging

Deliverable (end of Week 10) [Must be Submitted on Muele on Friday, 1ST May, 2026]

1. **Final Capstone Portfolio (single PDF document, 15–25 pages)** including:
 - project context + stakeholder map
 - ethics charter + obligations map
 - harm register + mitigations
 - privacy/harassment controversy handling
 - IP & OSS compliance
 - cyberlaw/security pack + incident mini-plan
 - ISO 12207 mapping + gap improvement
 - ISO 15288 systems view + V&V excerpt
 - ISO 25000 quality plan + RTM excerpt + glossary
 - standards relationship map + reflection (what changed in project because of ethics/standards)
2. **ClickUp Export Evidence:** tasks per member across 10 weeks + completion status
3. **Final Presentation (Powerpoint) (20 minutes) and Q&A readiness**

Weekly Reporting Template (mandatory)

Each week, submit a 1-page report on your group Clickup space containing:

- Group name + project name + ClickUp link

- Member task list (who did what) + outcome summary
- Evidence attached/linked
- Risks/issues + mitigation
- Plan for next week